Karol Dawid Penczek

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Education

- **Mickiewicz High School No. 3** — Katowice, Poland
  International Baccalaureate Programme — *September 2019 - May 2023*
  Higher Level subjects: *English B, Mathematics AA HL, Computer Science*
  IB Score: 39/45

- **Eindhoven University of Technology** — Eindhoven, Netherlands
  Bachelor of Computer Science and Engineering — *September 2023 - Present*
  GPA 8.3/10
  Participant in the Student Teams programme

Experience

- **Aster in Space - Student Team** — Eindhoven, Netherlands
  Software Engineer — *September 2024 - December 2025*
  - **AetherSense Project (September 2024 - May 2025)**: Software Subteam Member
    - Implemented flight software for the successfully launched AetherSense scientific payload.
    - Developed embedded software by converting mathematical models into onboard code, including communication routines, data processing pipelines, sensor scanning logic, and attitude determination algorithms.
    - Conducted exploratory and design work such as comparing microcontroller options, assessing software environments, and defining system-level and software-level requirements.
    - Collaborated closely with mechanical, electronics, and systems engineers in a multidisciplinary team.
    - Worked under strict constraints on computational resources, power, and memory.
  - **CubeSat Project (June 2025 - December 2025)**: Software Subteam Member
    - Performed exploratory and design-oriented work by choosing the right software environments and hardware for the 3U Satellite.
    - Developed requirements for the software in collaboration with the other subteams.
    - Worked on implementing the project's software in accordance with the defined requirements.

- **CNN Concept Learning: Bachelor End Project** — Eindhoven, Netherlands
  Student — *February 2026 - Present*
  - **Monosemantic Concept Disentanglement Pipeline**:
    - Developed a pipeline combining CNNs and Sparse Autoencoders (SAEs) to disentangle internal representations into monosemantic concepts (e.g., "cat ear").
    - Applied sparse latent representations to interpret learned CNN features.
    - Investigating anchor-based supervision on the APT-36K dataset to improve concept separability and interpretability.

- **ML Collective** — Remote
  Member of an independent research organisation — *October 2024 - Present*
  - **Dragonfly Project (October 2025 - Present)**: Team member
    - Researched datasets and models that can be used for the project as well as related academic literature.
    - Together with the team developed adversarial cloaking and data poisoning methods for audio ML models, using CLAP and AudioLDM latent spaces to disrupt recognition while preserving perceptual quality.
    - Tuned the LoRAW model on the MTG-Jamendo dataset for evaluation and testing.
    - Repository: https://github.com/damanimc/Dragonfly
  - **Self-organised Study**:
    - Participated in the organisation's Research Jams and paper readings.
    - Studied recommended resources such as fast.ai, *Hands-On Graph Neural Networks Using Python*, and Andrej Karpathy's *Neural Networks: Zero to Hero* series.

- **Course 2IRR00: Software Design** — Eindhoven, Netherlands
  Teaching Assistant — *April 2025 - June 2025, Reappointed April 2026 - June 2026*
  - **Selection and scope**: Chosen as Teaching Assistant after completing the course with a top grade. Supported instruction in a large course with over 400 students and acted as the primary contact for approximately 60 students in a tutorial group.
  - **Tutorial sessions and project support**: Supervised student project groups, led Q&A and project demonstration sessions, and clarified course material during weekly tutorial meetings.
  - **Examinations**: Supervised and graded midterm examinations, ensuring consistent and fair application of the grading rubric.

- **Soft skill development**: Strengthened communication, mentoring, feedback, classroom management, and leadership skills by guiding diverse student teams and supporting them during project work.
- **Software design expertise**: Deepened understanding of software design principles (e.g. modularisation, abstraction, separation of concerns, basic design patterns) by reviewing student designs and discussing alternative architectures and trade-offs.

Multidisciplinary CBL Course Eindhoven, Netherlands
Student *April 2025 - June 2025*

- **Spatio-temporal pedestrian flow prediction model**:
  - In a team project, individually designed and implemented a spatio-temporal Graph Neural Network (GNN) with a GCN-based architecture to model and predict pedestrian flows in a busy train station environment.
  - Used **PyTorch Geometric** to construct dynamic graphs where nodes represent pedestrians, doors, and static obstacles, and edges encode spatial proximity and interactions over time.
  - Designed a data transformation pipeline converting raw trajectory coordinates into time-indexed graph sequences with node features, edge indices, and rolling prediction windows.
  - Implemented sequence DataLoaders and an autoregressive rollout procedure to forecast multiple future time steps and generate realistic crowd-movement trajectories.
  - Added evaluation and visualisation utilities to compare predicted vs. ground-truth trajectories and to inspect learned interaction patterns in the graph.
  - Collaborated with teammates on the literature review, project report, and development of an alternative baseline model.
  - Repository: https://github.com/karolos1444/FlowGroup1

Software Engineering Project Course: Facial Recognition Model for PeerCode Eindhoven, Netherlands
Student *September 2025 - January 2026*

- **End-to-end facial recognition system**:
  - Collaborated with an industry partner (PeerCode) to build a facial recognition service that processes both images and short video clips.
  - Packaged the backend as Docker containers exposing APIs for face detection, identification, and verification, along with a React-based web interface.
  - Assembled a synthetic (GDPR-compliant) gallery of identities and implemented an input pipeline for face localisation, alignment, and normalisation before inference.
  - Integrated and evaluated a deep learning model for face matching, systematically testing behaviour under varied lighting, pose, resolution, and multi-person scenarios.
  - Designed verification and logging mechanisms based on image hashing and embedding similarity scores to support controlled and secure system usage.

Other Projects

- **Passion-driven ML model repository**: Maintaining a personal GitHub repository of small AI models and experiments implemented from scratch to deepen understanding of core machine learning concepts (github.com/karolos1444/ML-projects).

- **SQL + PHP database for local internet provider**: For the Computer Science internal assessment (20% of the final IB score), created a customer record database using SQL and PHP.

Skills Summary

- **Languages**: English (C1), Polish (native), German (B2)
- **Programming Languages**: Python, PHP, C, C++, JavaScript, SQL, Java, Assembly, Rust, Haskell
- **Frameworks & Tools**: React.js, PyTorch, Docker, Streamlit, FreeRTOS
- **Soft Skills**: Leadership, Project & team management, Technical writing, Communication